

RESEARCH VISION

My research vision is to build **provably safe, verifiable, and trustworthy AI systems**. I bridge formal language theory and modern AI/NLP, applying theoretical tools—formal grammars and automata—to create AI systems with **mathematical guarantees** on safety and correctness.

- **Core Approach: Bridging Formal Language Theory and Modern AI**

My Ph.D. research bridges formal language theory and AI/NLP to address this challenge. I advance foundational questions in computational theory—such as decidability of string properties (Parikh matrices), language hierarchies, and formal characterization of structural constraints—and directly apply these theoretical tools to build **verifiable AI systems**. By integrating symbolic, grammar-based methods with neural models, I create frameworks where AI behavior can be formally specified and mathematically verified.

- **Future Focus: Verifiable LLM Safety**

I aim to apply these methodologies to the most pressing AI challenge: **provably safe LLMs**. By formalizing safety constraints as decidable properties in formal language theory, I will develop systems with **mathematical verification** of outputs, guaranteed abstention mechanisms, and transparent specifications that enable external auditing. My unique expertise in both formal methods and modern NLP positions me to pioneer **verifiable AI safety**—moving beyond heuristics to mathematical guarantees.

EDUCATION

Yonsei University

Integrated Ph.D. course in Computer Science
Supervisor: Yo-Sub Han

Seoul, KR

2019.03–2026.02

Thesis: *Symbolic Techniques for Deep Learning in Natural Language Processing and Program Analysis*

Yonsei University

Bachelor's degree in Computer Science

Seoul, KR

2015.03–2019.02

SELECTED PUBLICATIONS (VERIFICATION & FORMAL METHODS)

- **Query4Regex: Verifiable Regex Transformation through Formal Operations from NL and DSL Queries** (Findings of EACL 2026).
Verifiable regular expression (regex) transformation with LLMs: LLMs transform regexes executing formal operators given in a query. The work checks semantic accuracy via automata equivalence.
- **Repairing Regex Vulnerabilities via Localization-Guided Instructions** (EACL 2026).
LLM-assisted repair constrained by symbolic vulnerability localization; reliability-oriented program repair for security-critical regex artifacts.
- **TCProF: Time-Complexity Prediction SSL Framework** (NAACL 2025).
Neuro-symbolic semi-supervised learning for program analysis with symbolic guidance (structure-aware pseudo-labeling and co-training).
- **Characterizations of M -equivalence and weak M -relation** (IJFCS 2025).
Foundations for equivalence and rewriting transformation rules on words via Parikh matrices; formal characterization of M -equivalence and rule-based transformations.

Peer-Reviewed Conference/Journal Papers

- [C1] **Query4Regex: Verifiable Regex Transformation through Formal Operations from NL and DSL Queries**
Joonghyuk Hahn, Yo-Sub Han
In Findings of EACL 2026 [\[PDF\]](#)
- [C2] **Repairing Regex Vulnerabilities via Localization-Guided Instructions**
Sicheol Sung[†], Joonghyuk Hahn[†], Yo-Sub Han
In Proceedings of EACL 2026 [\[PDF\]](#)
- [C3] **A Regex Minimization Benchmark: A PSPACE-Complete Challenge for Language Models**
Hyundong Jin, Joonghyuk Hahn, Yo-Sub Han
In Proceedings of EACL 2026 [\[PDF\]](#)
- [C4] **EnCur: Curriculum-Based In-Context Learning with Structural Encoding for Code Time Complexity Prediction**
Joonghyuk Hahn, Aditi, Seung-Yeop Baik, Shinwoo Park, Sang-Ki Ko, Yo-Sub Han
To appear in ESWA 2026 [\[PDF\]](#)
- [C5] **AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection**
Yejin Lee, Joonghyuk Hahn, Hyeseon Ahn, Yo-Sub Han
In Proceedings of EMNLP 2025 [\[PDF\]](#)
- [C6] **CodeComplex: Dataset for Worst-Case Time Complexity Prediction**
Seung-Yeop Baik, Joonghyuk Hahn, Jungin Kim, Mingi Jeon, Aditi, Yo-Sub Han, Sang-Ki Ko
In Findings of EMNLP 2025 [\[PDF\]](#)
- [C7] **TCProF: Time-Complexity Prediction SSL Framework**
Joonghyuk Hahn, Hyeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han
In Proceedings of NAACL 2025 [\[PDF\]](#)
- [C8] **Advanced Code Time Complexity Prediction Approach Using Contrastive Learning**
Shinwoo Park, Joonghyuk Hahn, Elizabeth Orwig, Sang-Ki Ko, Yo-Sub Han
In EAAI 2025, Vol. 151 [\[PDF\]](#)
- [C9] **Characterizations of M-equivalence and weak M-relation**
Joonghyuk Hahn, Hyunjoon Cheon, Yo-Sub Han
In IJFCS 2025 [\[PDF\]](#)
- [C10] **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], Joonghyuk Hahn[†], Yo-Sub Han
In Theory of Computing Systems 2024, Vol. 68, 301–321 [\[PDF\]](#)
- [C11] **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem**
Ingyu Baek[†], Joonghyuk Hahn[†], Yo-Sub Han, Kai Salomaa
In Proceedings of DLT 2024, LNCS 14791, 68–81 [\[PDF\]](#)
- [C12] **SuperST: Superficial Self-Training for Few-Shot Text Classification**
Ju-Hyoung Lee[†], Joonghyuk Hahn[†], Hyeon-Tae Seo, Jiho Park, Yo-Sub Han
In Proceedings of LREC-COLING 2024 [\[PDF\]](#)
- [C13] **ATHENA: Mathematical Reasoning with Thought Expansion**
J.B. Kim, Hazel Kim, Joonghyuk Hahn, Yo-Sub Han
In Proceedings of EMNLP 2023, 16315–16327 [\[PDF\]](#)
- [C14] **GDA: Grammar-based Data Augmentation for Text Classification using Slot Information**
Joonghyuk Hahn, Hyunjoon Cheon, Elizabeth G. Orwig, Su-Hyeon Kim, Sang-Ki Ko, Yo-Sub Han
In Findings of EMNLP 2023 [\[PDF\]](#)
- [C15] **M-equivalence of Parikh Matrix over a Ternary Alphabet**
Joonghyuk Hahn, Hyunjoon Cheon, Yo-Sub Han
In Proceedings of CIAA 2023 [\[PDF\]](#)

- [C16] **Boosting Code Summarization by Embedding Code Structures**
Jikyeong Son[†], **Joonghyuk Hahn[†]**, Hyeon-Tae Seo, Yo-Sub Han
In Proceedings of COLING 2022, 5966–5977 [PDF]
- [C17] **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], **Joonghyuk Hahn[†]**, Yo-Sub Han
In Proceedings of DLT 2022, LNCS 13257, 115–126 [PDF]
- [C18] **Self-Training using Rules of Grammar for Few-Shot NLU**
Joonghyuk Hahn, Hyunjoon Cheon, Kyuyeol Han, Cheongjae Lee, Junseok Kim, Yo-Sub Han
In Findings of EMNLP 2021 [PDF]
- [C19] **Most Pseudo-copy Languages Are Not Context-Free**
Hyunjoon Cheon[†], **Joonghyuk Hahn[†]**, Yo-Sub Han, Sang-Ki Ko
In Proceedings of COCOON 2021, 189–200 [PDF]

Under Review

- **MEC³O: Multi-Expert Consensus for Code Time Complexity Prediction**
Joonghyuk Hahn[†], Soohan Lim[†], Yo-Sub Han
Under review [PDF]
- **ECO: Enhanced Code Optimization via Performance-Aware Prompting for Code-LLMs**
Su-Hyeon Kim, Joonghyuk Hahn, Sooyung Cha, Yo-Sub Han
Under review [PDF]
- **A Voronoi-Embedding Framework for Semi-Supervised Time-Complexity Prediction**
Hyeseon Ahn, Joonghyuk Hahn, Yo-Sub Han
Under review
- **ContractEval: A Benchmark for Evaluating Contract-Satisfying Assertions in Code Generation**
Soohan Lim[†], Joonghyuk Hahn[†], Yo-Sub Han
Under review [PDF]
- **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem (Journal Extension)**
Ingyu Baek[†], Joonghyuk Hahn[†], Yo-Sub Han, Kai Salomaa
Under review

PROJECTS

2023–2025: Human–AI Programming Platform Research (Follow-up)

Ministry of Science and ICT (MSIT), Republic of Korea

- Studied Human-in-the-Loop workflows using large language models (LLMs) to infer developers’ programming intent and generate reliable source code.
- Investigated prompt engineering techniques to control hallucinations during code analysis and improve consistency of LLM-assisted programming.

2023–2024: Deep Learning-based Financial Data Analysis & Transformation

Bankware Global

- Led a data-engineering methodology to analyze specialized financial data and transform it into LLM-friendly formats for downstream reasoning and retrieval.
- Explored domain adaptation directions for finance-specialized models using limited expert data, including instruction tuning and retrieval-augmented generation (RAG) to improve performance.

2022: 5th AI Grand Challenge

Ministry of Science and ICT (MSIT), Republic of Korea

- Developed deep learning models for inferring solution processes and final answers for descriptive (free-form) math problems; participated in the national-scale competition finals.
- Designed equation-based heuristics to generate Korean training data; fine-tuned language models for mathematical problem solving and code-like reasoning.

- Passed the first stage and drafted the project proposal; then participated in the second stage.

2019–2022: Human–AI Collaborative Programming Platform Research

Ministry of Science and ICT (MSIT), Republic of Korea

- Designed and studied platform frameworks for efficient collaboration between humans and AI in programming workflows.
- Developed neuro-symbolic and semi-supervised learning ideas for reliable code-related tasks (e.g., code summarization and complexity-oriented analysis) under limited supervision.

2019–2022: Automated Malware Pattern Rule-set Generation

Ministry of Science and ICT (MSIT), Republic of Korea

- Developed automated pipelines to generate malware pattern rule-sets by training deep learning models for malware pattern detection.
- Studied rule-set compression and synthesis from learned signals to support scalable detection and automation.

RESEARCH INTERESTS FOR FUTURE WORKS

- **Mechanistic Interpretability of Generative Models:** Applying formal language theory and automata to understand internal mechanisms of LLMs—analyzing learned circuits, feature representations, and reasoning processes to make models more transparent and interpretable.
- **Formal Reasoning & Symbolic Integration:** Bridging symbolic methods (formal grammars, logical constraints, decidability analysis) with neural models to enable verifiable reasoning, compositional generalization, and systematic problem-solving in language models.
- **Constrained & Controllable Generation:** Developing grammar-based frameworks for structured, safe generation with guarantees on output properties—applicable to code synthesis, mathematical reasoning, and domain-specific language generation.
- **Efficient & Robust Language Models:** Investigating byte-level models, robustness to perturbations, and efficient architectures informed by formal language hierarchies—advancing foundations for more resilient and efficient generative systems.
- **Neuro-Symbolic AI Safety:** Bridging symbolic reasoning (formal methods, logical constraints) with neural models to create interpretable, verifiable AI systems with transparent failure modes.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java
- **ML/NLP Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers
- **Formal Methods:** Automata theory, formal grammars, decidability analysis, formal language hierarchies
- **Research Areas:** constrained generation, neuro-symbolic AI, formal verification for neural networks
- **Tools:** L^AT_EX, Git, Linux/Unix

REFERENCES

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